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# Tools, Not Talk: The Knowledge Ceiling of Biology Agents

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

1        Agentic systems for biology are increasingly built around orchestration: critics  
2        that review answers, teams of role-playing scientists that debate, and models that  
3        reason for longer. We show that on objectively scored biology research tasks this  
4        machinery buys almost nothing, and we identify why. Across 150 pre-registered  
5        LAB-Bench and LABBench2 questions, compared at a matched budget of three  
6        model calls, self-critique never beats simple majority voting, multi-agent debate  
7        helps only a model whose reasoning is switched off, and a 25-fold increase in  
8        generated tokens from longer reasoning leaves accuracy unchanged. The reason  
9        is a ceiling set by the questions themselves: which question is asked explains 56%  
10       of the variance in correctness, while the choice of model, strategy and reasoning  
11       effort explains 0.6%. That ceiling is missing knowledge, not missing deliberation.  
12       A single model call with web search and code execution solves 47% of the ques-  
13       tions that every orchestrated and reasoning-heavy system failed. For life-science  
14       agents, the budget belongs in tools, not talk.

## 15    1 Introduction

16    LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology  
17    [1, 2, 5, 6, 8, 16]. Many of these systems are organised as conversations: a critic reviews a draft,  
18    several specialised scientists debate, an adjudicator decides. Their gains are usually reported against  
19    a single model call, even though the agent makes many. Sampling several answers and voting already  
20    improves accuracy [3, 12, 19], so these comparisons cannot tell whether the orchestration helps or  
21    whether the extra calls do [9].

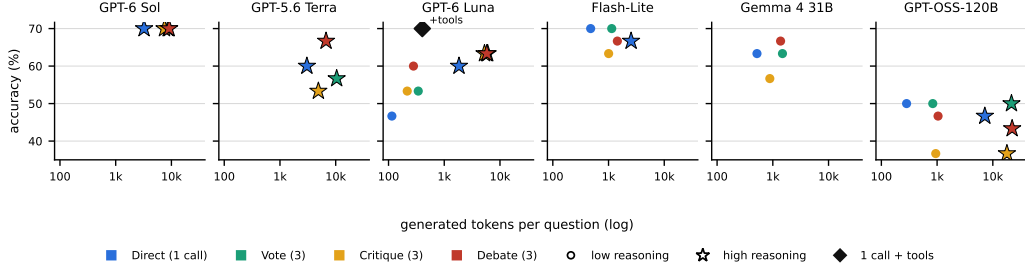
22    We separate the two. Every orchestrated strategy receives exactly the budget of a voting baseline:  
23    three calls, the same model, the same prompts and the same decoding. We then ask what actually  
24    moves accuracy on biology research tasks: orchestration, longer reasoning, or access to knowledge  
25    through tools.

## 26    Contributions.

- 27        • **Orchestration buys little at matched compute.** Across three models, self-critique never  
28        beats voting and repairs no more answers than it breaks. Debate helps only when the  
29        model cannot reason on its own. Raising reasoning effort, for six model families, rarely  
30        helps either (§3.1–3.2).
- 31        • **The question decides.** Every system solves the same questions and fails the same ques-  
32        tions. Question identity explains 56% of the variance in correctness and the system 0.6%  
33        (§3.3).
- 34        • **Tools break the ceiling.** One tool-using call solves 47% of the questions that no other  
35        configuration solved, and more compute on top of tools adds comparatively little (§3.4).

**Table 1:** Accuracy (%) at a matched budget of three calls (DIRECT uses one).  $\Delta$  = DEBATE − VOTE in points, with paired bootstrap 95% CI.

| Model                     | $n$ | DIRECT | VOTE | CRITIQUE | DEBATE      | $\Delta$ [95% CI]         |
|---------------------------|-----|--------|------|----------|-------------|---------------------------|
| GPT-OSS-120B              | 150 | 46.7   | 49.3 | 42.7     | 47.3        | -2.0 [-8.7, +4.7]         |
| Gemma 4 31B (no thinking) | 150 | 48.7   | 48.7 | 47.3     | <b>57.3</b> | <b>+8.7</b> [+2.0, +15.3] |
| Flash-Lite                | 75  | 61.3   | 62.7 | 56.0     | 57.3        | -5.3 [-13.3, +1.3]        |



**Figure 1: Every way of spending inference lands on the same per-model plateau; tools rise above it.** Accuracy vs. generated tokens per question on a shared 30-question set. Colour marks the strategy; circles are low reasoning effort and stars high. Across nearly two orders of magnitude in tokens, accuracy barely moves within a model. The black diamond, one GPT-6 Luna call with web search and code execution, is the only configuration above its plateau, and it is among the cheapest.

## 2 Setup

**Tasks.** We draw 30 questions from each of five categories (150 in total): four from LAB-Bench [10] (protocol troubleshooting, sequence manipulation, database and literature questions; multiple choice) and open-ended sequence computations from LABBench2 SeqQA2 [11], scored with the official validators. All answers are scored deterministically, with no LLM judge.

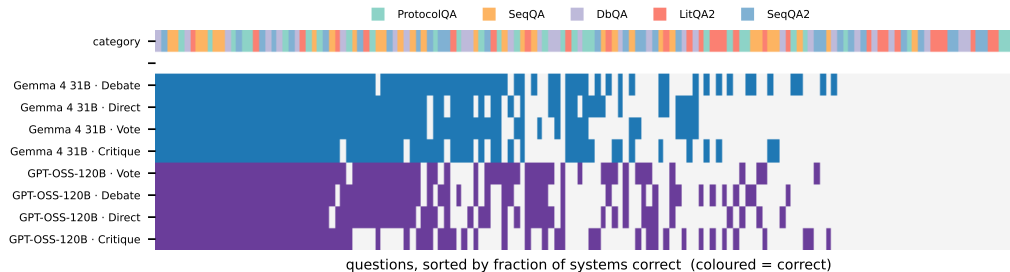
**Strategies.** DIRECT makes one call. VOTE makes three independent calls and takes the majority answer. CRITIQUE solves, critiques its own solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve independently, then an adjudicator decides. VOTE, CRITIQUE and DEBATE each use exactly three calls, and only visible outputs pass between calls (prompts in App. A).

**Models.** The matched-budget study uses GPT-OSS-120B [13] (low reasoning effort), Gemma 4 31B [17] (thinking disabled) and Gemini 3.5 Flash-Lite (low thinking) on all 150 questions (Flash-Lite on a fixed 75). To test reasoning effort and tools, we add GPT-6 Luna, GPT-6 Sol and GPT-5.6 Terra, and rerun GPT-OSS and Flash-Lite at high effort, on a fixed 30-question subset with 6 per category. Task IDs, prompts, decoding and the primary analysis were frozen before evaluation (App. B). All contrasts are paired over questions, with bootstrap 95% CIs and exact McNemar tests.

## 3 Results

### 3.1 Orchestration buys little at matched compute

Three calls spent on orchestration do no better than three calls spent on independent samples (Table 1). **Self-critique never beats voting** for any model (−6.7, −1.3, −6.7 points; pooled −4.5 [−9.3, +0.3]). It breaks correct answers as readily as it fixes wrong ones: relative to the vote it rescues 24%, 13% and 25% of errors but corrupts 38%, 16% and 26% of correct answers. The damage is worst on exact sequence manipulation (SeqQA: -23 and -20 points for GPT-OSS and Flash-Lite), where a fluent but wrong critique argues the model out of a correct computation. **Debate helps only the model that cannot think.** For Gemma, whose thinking is disabled, debate beats voting by +8.7 points ( $p = 0.015$ ); for the two models that reason, it does not (−2.0 and −5.3).



**Figure 2: The question, not the system, decides.** Correctness of every model × strategy system (rows) on every question (columns, sorted by the fraction of systems correct). All systems solve and fail the same questions. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

### 3.2 Longer reasoning does not substitute for knowledge

Raising reasoning effort multiplies the cost without moving accuracy (Fig. 1). GPT-OSS at high effort generates  $25\times$  more tokens and scores 47% against 50%; Flash-Lite spends  $5\times$  more tokens and scores 67% against 70%. The one model that gains is GPT-6 Luna, which uses no reasoning tokens at low effort, and it gains only up to the level its low-effort debate had already reached (60% vs. 60%). Debate and reasoning are therefore substitutes, and neither gets past the plateau. Stacking orchestration on high effort adds nothing: GPT-6 Sol scores 70% with every strategy while the three-call strategies generate  $\sim 3\times$  the tokens.

### 3.3 The question decides

A two-way decomposition of the correctness matrix attributes 56% of the variance to the question and 0.6% to the system (Fig. 2). The pattern holds across all 33 model, strategy and effort configurations on the shared subset, frontier models at high effort included (47% vs. 3.9%). Every system solves the same easy questions and fails the same hard ones. The 32 questions that no system solves are dominated by literature and database recall (18) and open-ended sequence computation (9). These are failures of knowledge, which no amount of deliberation can supply.

### 3.4 Tools break the ceiling

Giving GPT-6 Luna web search and code execution in a single call raises accuracy on the shared subset from 47% to 70% (+23 points [+7, +43],  $p = 0.039$ ). That is more than any orchestration or reasoning setting achieves, at  $\sim 404$  generated tokens per question. On the 32 questions that every orchestrated and reasoning-heavy system failed, the same single call reaches 47%, against 6% without tools (+41 points [+25, +56],  $p = 0.0002$ ). Literature questions rise from 18% to 91% and database questions from 0% to 43%. Retrieval helping on these categories is expected, since they were designed for retrieval agents [10, 14]. The point is the contrast: no orchestration or reasoning setting moved these questions at all. Compute on top of tools adds comparatively little. High effort reaches 83% (+13,  $p = 0.22$ ) and a three-call vote 77% (+7,  $p = 0.73$ ), at  $\sim 6\times$  and  $\sim 3.5\times$  the tokens. Tool calls never reached the benchmark or its datasets.

## 4 Implications for building life-science agents

**Benchmark against a matched vote.** A multi-call agent should beat independent sampling with the same number of calls, not a single call [9]. Most orchestration gains disappear under this baseline, consistent with compute-matched studies of debate in general reasoning [4, 15, 18, 20] and with the unreliability of intrinsic self-correction [7]. **Spend the budget on tools, not talk.** Critics, debaters and longer chains of thought re-deliberate over knowledge the model does not have. Retrieval supplies it, at a fraction of the tokens. **Build domain tools for computation.** A generic shell barely helps open-ended sequence design and analysis (0% to 11%). The next gains lie in biology-specific tool ecosystems and specialist models, not in more conversation between copies of the same model.

98 **Scope.** Our tasks have deterministic answers, and we study short-horizon question answering  
 99 rather than long-running discovery loops. The frontier and tool comparisons use a 30-question  
 100 subset, and the tool condition adds a one-sentence note announcing the tools.

## 101 5 Conclusion

102 On biology research tasks, how a model spends extra inference barely changes which questions it  
 103 gets right. This holds for voting, critique, debate and longer reasoning alike: the question explains  
 104 56% of the variance in correctness and the system 0.6%. The ceiling is knowledge, and a single  
 105 tool-using call breaks it where orchestration does not. Life-science agents should spend their budget  
 106 on tools, not talk.

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## A Prompts

All prompts are single user messages. {QUESTION} is the question (with protocol, options or injected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official SeqQA2 suffix.

### Direct / Independent×3 / Critique call 1 (solve)

Solve the following biology research question.

{QUESTION}

{ANSWER INSTRUCTION}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

### Critique call 2 (critique)

Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which answer you believe is correct and why. If you find no error, say so.

{QUESTION}

Proposed solution:

{CALL 1 OUTPUT}

Respond with at most four sentences, starting with "CRITIQUE:".

200 **Critique call 3 (revise)**

201 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that  
202 attempt. Either may contain mistakes. Considering both, produce the final answer.

203  
204 {QUESTION}

205  
206 First attempt:

207 {CALL 1 OUTPUT}

208  
209 Reviewer's critique:

210 {CALL 2 OUTPUT}

211  
212 {ANSWER INSTRUCTION}

213  
214 Respond in exactly this format and keep it short:

215 <answer>YOUR ANSWER</answer>

216 CONFIDENCE: <probability between 0 and 1 that your answer is correct>

217 JUSTIFICATION: <at most two sentences>

218 **Debate call 1 (researcher)**

219 You are a molecular biology researcher. Solve the following biology research question independently.

220  
221 {QUESTION}

222  
223 {ANSWER INSTRUCTION}

224  
225 Respond in exactly this format and keep it short:

226 <answer>YOUR ANSWER</answer>

227 CONFIDENCE: <probability between 0 and 1 that your answer is correct>

228 JUSTIFICATION: <at most two sentences>

229 **Debate call 2 (skeptic)**

230 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking  
231 especially for subtle experimental, biological or computational errors and traps.

232  
233 {QUESTION}

234  
235 {ANSWER INSTRUCTION}

236  
237 Respond in exactly this format and keep it short:

238 <answer>YOUR ANSWER</answer>

239 CONFIDENCE: <probability between 0 and 1 that your answer is correct>

240 JUSTIFICATION: <at most two sentences>

241 **Debate call 3 (adjudicator)**

242 Two scientists independently analysed the biology research question below. Adjudicate between their analyses  
243 (either or both may be wrong) and submit the final answer.

244  
245 {QUESTION}

246  
247 Analysis 1 (molecular biology researcher):

248 {CALL 1 OUTPUT}

249  
250 Analysis 2 (skeptical scientific reviewer):

251 {CALL 2 OUTPUT}

252  
253 {ANSWER INSTRUCTION}

254  
255 Respond in exactly this format and keep it short:

256 <answer>YOUR ANSWER</answer>

257 CONFIDENCE: <probability between 0 and 1 that your answer is correct>

258 JUSTIFICATION: <at most two sentences>

## B Pre-registration deviations

All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS questions served in full by a second provider of the same model (identical prompts, seeds and cap; excluding them leaves DEBATE – VOTE at  $-1.6$ ), with partially served questions discarded and re-run on the primary provider; (iii) the reasoning-effort and tool arms on the fixed 30-question subset and the unsolved set, added after the freeze. Frontier models were served through their vendor CLI with web search, code execution and sub-agents disabled except in the tool condition (a first attempt with web search left on was discarded unanalysed); GPT-OSS at high effort was served by a second host whose low-effort accuracy matches the main run. The tool condition prepends one sentence announcing the tools, and all searches and commands were logged and checked for benchmark access (none).

## C Additional results

| Category   | Gemma |      |      |      |          | Lite |      |      |      |          | OSS  |      |      |      |          |
|------------|-------|------|------|------|----------|------|------|------|------|----------|------|------|------|------|----------|
|            | D     | I    | C    | M    | Disagree | D    | I    | C    | M    | Disagree | D    | I    | C    | M    | Disagree |
| DbQA       | 56.7  | 53.3 | 40.0 | 56.7 | 33.3     | 40.0 | 40.0 | 40.0 | 33.3 | 26.7     | 46.7 | 46.7 | 43.3 | 36.7 | 36.7     |
| LitQA2     | 33.3  | 30.0 | 33.3 | 36.7 | 46.7     | 60.0 | 53.3 | 53.3 | 46.7 | 26.7     | 20.0 | 30.0 | 40.0 | 36.7 | 43.3     |
| ProtocolQA | 56.7  | 56.7 | 60.0 | 70.0 | 20.0     | 66.7 | 80.0 | 86.7 | 73.3 | 26.7     | 56.7 | 50.0 | 43.3 | 46.7 | 33.3     |
| SeqQA      | 56.7  | 56.7 | 66.7 | 73.3 | 46.7     | 73.3 | 73.3 | 53.3 | 66.7 | 13.3     | 63.3 | 66.7 | 43.3 | 60.0 | 33.3     |
| SeqQA2     | 40.0  | 46.7 | 36.7 | 50.0 | 63.3     | 66.7 | 66.7 | 46.7 | 66.7 | 40.0     | 46.7 | 53.3 | 43.3 | 56.7 | 43.3     |

**Table 2:** Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.